Conditional simulation of the original fitted GP

Fitted model and exact coordinate union

Same kernel, mean, parameters and noise as model selection

$U_d = \mathrm{unique}(X_{:,d} \cup X_{*,:,d});\ U = U_1 \times \cdots \times U_D$

Separable covariance terms

$$K_{uu} = \sum_{r=1}^{R} \bigotimes_{d=1}^{D} K^{(r,d)}$$

$B_{r,d} B_{r,d}^T = K^{(r,d)}$

Nonuniform Cartesian axes are allowed

Original GP conditioning matrix

$A_{\mathrm{ex}} = K_{XX} + \sigma_n^2 I_n = LL^T$

$P_X,\ P_*$: exact node selectors

No inducing interpolation

Independent prior contributions

$$u_0 = \sum_r (\bigotimes_d B_{r,d})\, \xi_r$$

Independent $\xi_r \sim \mathcal{N}(0, I_m)$

Independent $\varepsilon_0 \sim \mathcal{N}(0, \sigma_n^2 I_n)$

Reusable training-system solve

$r_y = y - \mu_X - P_X u_0 - \varepsilon_0$

$LL^T a = r_y$

One $n \times n$ factor reused per draw

Condition the joint draw at the query nodes

$f_* = \mu_* + P_* u_0 + K_{*X} a$

No added query observation noise; no $q \times q$ covariance factorization

n: observations; m: union nodes; q: query nodes; R: separable terms

Original-GP identity; factor roundoff and Monte Carlo agreement are checked separately.